

Proof. It follows from the closed-form expression of MMD distance that we have

$$\begin{aligned}
\text{MMD}^2\left((f_{r,\gamma})_{\#}\mu, (f_{r,\gamma})_{\#}\nu; k\right) &= \int \int k(z, z')(f_{r,\gamma})_{\#}\mu(dz)(f_{r,\gamma})_{\#}\mu(dz') + \int \int k(w, w')(f_{r,\gamma})_{\#}\nu(dw)(f_{r,\gamma})_{\#}\nu(dw') \\
&\quad - 2 \int \int k(z, w)(f_{r,\gamma})_{\#}\mu(dz)(f_{r,\gamma})_{\#}\nu(dw) \\
&= \int \int k(r + \gamma z, r + \gamma z')\mu(dz)\mu(dz') + \int \int k(w, r + \gamma w')\nu(dw)\nu(dw') \\
&\quad - 2 \int \int k(r + \gamma z, r + \gamma w)\mu(dz)\nu(dw) \\
&= |\gamma|^\alpha \int \int k(z, z')\mu(dz)\mu(dz') + |\gamma|^\alpha \int \int k(w, w')\nu(dw)\nu(dw') \\
&\quad - 2|\gamma|^\alpha \int \int k(z, w)\mu(dz)\nu(dw) \\
&= |\gamma|^\alpha \text{MMD}^2(\mu, \nu; k).
\end{aligned}$$

□

Lemma 5. For any $\mu, \nu \in \mathcal{P}(\mathcal{X})$, $(\beta_i)_{i \in I} \subset \mathbb{R}$ for some indices I , we have

$$\text{MMD}^2(\mu, \nu; \sum_{i \in I} \beta_i k_i) = \sum_{i \in I} \beta_i \text{MMD}^2(\mu, \nu; k_i)$$

Proof. We have

$$\begin{aligned}
\text{MMD}^2(\mu, \nu; \sum_{i \in I} \beta_i k_i) &= \int \int \sum_{i \in I} \beta_i k_i(z, z')\mu(dz)\mu(dz') + \int \int \sum_{i \in I} \beta_i k_i(w, w')\nu(dw)\nu(dw') - 2 \int \int \sum_{i \in I} \beta_i k_i(z, w)\mu(dz)\nu(dw) \\
&= \sum_{i \in I} \beta_i \left(\int \int k_i(z, z')\mu(dz)\mu(dz') + \int \int k_i(w, w')\nu(dw)\nu(dw') - 2 \int \int k_i(z, w)\mu(dz)\nu(dw) \right) \\
&= \sum_{i \in I} \beta_i \text{MMD}^2(\mu, \nu; k_i).
\end{aligned}$$

□

Lemma 6. If k is shift invariant and scale sensitive with order $\alpha > 0$, then

$$\text{MMD}_\infty(\mathcal{T}^\pi \mu, \mathcal{T}^\pi \nu; k) \leq \gamma^{\alpha/2} \text{MMD}_\infty(\mu, \nu; k),$$

for any $\mu, \nu \in \mathcal{P}(\mathcal{X})$ and any (stationary) policy π .

Proof. We have

$$\begin{aligned}
&\text{MMD}^2\left(\mathcal{T}^\pi \mu(s, a), \mathcal{T}^\pi \nu(s, a); k\right) \\
&= \text{MMD}^2\left(\int (f_{r,\gamma})_{\#}\mu(s', a')\pi(da'|s')P(ds'|s, a)\mathcal{R}(dr|s, a), \int (f_{r,\gamma})_{\#}\nu(s', a')\pi(da'|s')P(ds'|s, a)\mathcal{R}(dr|s, a); k\right) \\
&\stackrel{(a)}{\leq} \int \text{MMD}^2\left((f_{r,\gamma})_{\#}\mu(s', a'), (f_{r,\gamma})_{\#}\nu(s', a'); k\right)\pi(da'|s')P(ds'|s, a)\mathcal{R}(dr|s, a) \\
&\stackrel{(b)}{=} \gamma^\alpha \int \text{MMD}^2(\mu(s', a'), \nu(s', a'))\pi(da'|s')P(ds'|s, a)\mathcal{R}(dr|s, a) \\
&\leq \gamma^\alpha \text{MMD}_\infty^2(\mu, \nu).
\end{aligned}$$

Here (a) follows from Lemma 3, and (b) follows from Lemma 3. Finally, the last inequality concludes the lemma by the definition of supremum MMD. □

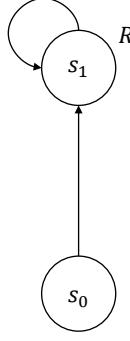


Figure 5: A simple MDP with 2 states: initial state s_0 and absorbing state s_1 . Any agent receives a reward $r \sim R$ whenever it reaches state s_1 .

Proof of Theorem 2.1. We have

$$\begin{aligned}
\text{MMD}^2\left(\mathcal{T}^\pi\mu(s, a), \mathcal{T}^\pi\nu(s, a); \sum_{i \in I} c_i k_i\right) &\stackrel{(a)}{=} \sum_{i \in I} c_i \text{MMD}^2\left(\mathcal{T}^\pi\mu(s, a), \mathcal{T}^\pi\nu(s, a); k_i\right) \\
&\stackrel{(b)}{\leq} \sum_{i \in I} c_i \gamma^{\alpha_i} \text{MMD}^2\left(\mu(s, a), \nu(s, a); k_i\right) \\
&\stackrel{(c)}{\leq} \gamma^{\alpha_*} \sum_{i \in I} c_i \text{MMD}^2\left(\mu(s, a), \nu(s, a); k_i\right) \\
&\stackrel{(d)}{=} \gamma^{\alpha_*} \text{MMD}^2\left(\mu(s, a), \nu(s, a); \sum_{i \in I} c_i k_i\right) \\
&\leq \gamma^{\alpha_*} \text{MMD}_\infty^2\left(\mu, \nu; \sum_{i \in I} c_i k_i\right).
\end{aligned}$$

Here, (a) and (d) follow from Lemma 5, (b) follows from Lemma 6, and (c) follows from that $\gamma \leq 1$ and $c_i \geq 0$. Finally, the last inequality above concludes the proof by the definition of supremum MMD. $\square$

Proof of Theorem 2.2

Proof of Theorem 2.2. Case 1: Gaussian kernels $k(x, y) = \exp(-(x - y)^2/(2\sigma^2))$.

We will prove that MMD_∞ associated with Gaussian kernels $k(x, y) = \exp(-(x - y)^2/(2\sigma^2))$ for some $\sigma > 0$ is not a contraction by contradiction and counterexamples. Assume by contradiction that there exists some $\alpha > 0$ such that

$$\text{MMD}_\infty(\mathcal{T}^\pi\mu, \mathcal{T}^\pi\nu; k) \leq \gamma^\alpha \text{MMD}_\infty(\mu, \nu; k), \quad (3)$$

for all $\mu, \nu \in \mathcal{P}(\mathcal{X})$. If $\alpha \geq 1$, $\gamma^\alpha \leq \gamma^{\alpha'}, \forall \alpha' \in (0, 1]$, thus without loss of generality, assume that $\alpha \in (0, 1]$.

We provide a counterexample that contradicts Inequality (3). Consider a simple MDP with only 2 states: initial state s_0 and absorbing state s_1 where an agent receives a reward $r \sim R$ whenever it reaches s_1 (see Figure 5). Assume that the reward distribution R has $\text{dom}(R) = \{r_1, \dots, r_n\}$ with respective probabilities $\{p_{r_1} = \epsilon, p_{r_2} = \epsilon, \dots, p_{r_{n-1}} = \epsilon, p_{r_n} = 1 - (n-1)\epsilon\}$ for some $\epsilon \in (0, \frac{1}{n-1})$ to be chosen later. Let $\mu, \nu \in \mathcal{P}(\text{dom}(R))^\mathcal{S}$ where $\mathcal{S} = \{s_0, s_1\}$ such that $\mu(s_0) = \mu(s_1) = p := \sum_{i=1}^n p_i \delta_{r_i}$ (for $\{p_i\}$ to be chosen later) and $\nu(s_0) = \nu(s_1) = q := \sum_{i=1}^n q_i \delta_{r_i}$ (for $\{q_i\}$ to be chosen later). It is easy to verify that for any policy π , we have

$$\begin{aligned}
\mathcal{T}^\pi\mu(s_0) &= \mathcal{T}^\pi\mu(s_1) = \mathbb{E}_{r \sim R} [(f_{r, \gamma})_\# p] \\
\mathcal{T}^\pi\nu(s_0) &= \mathcal{T}^\pi\nu(s_1) = \mathbb{E}_{r \sim R} [(f_{r, \gamma})_\# q],
\end{aligned}$$

where $f_{r, \gamma}(z) := r + \gamma z, \forall z$ and $\#$ denotes pushforward operation. Note that $(f_{r, \gamma})_\# \mu(s_1)$ assigns probabilities $\{p_1, \dots, p_n\}$ respectively to $r + \gamma \text{dom}(R) := \{r + \gamma r_1, \dots, r + \gamma r_n\}$, and similarly for $(f_{r, \gamma})_\# \nu(s_1)$. We have

$$\sum_{i=1}^n p_{r_i}^2 = (n-1)\epsilon^2 + (1 - (n-1)\epsilon)^2 \in [\frac{1}{n}, 1).$$

Parameters	Values
n	5
γ	0.8 (or 0.9)
σ	0.1
$\{p_i\}$	$[0.4, 0.3, 0.2, 0.1, 0]$
$\{q_i\}$	$[0, 0.1, 0.2, 0.3, 0.4]$
$\{r_i\}$	$[0, 0.25, 0.5, 0.75, 1.]$

Table 2: A realization of the parameters in the counterexample for proving Theorem 2.2

Note that since $\alpha \in (0, 1]$, we have $\gamma^{2\alpha} \in [\gamma^2, 1)$. Now, choose $\gamma \in (0, 1)$ such that $\gamma^2 \geq \frac{1}{n}$, then there exists $\epsilon \in (0, \frac{1}{n-1})$ such that $\sum_{i=1}^n p_{r_i}^2 = \gamma^{2\alpha}$. Define

$$\eta_r := (f_{r,\gamma})_{\#} p = \sum_{i=1}^n p_i \delta_{r+\gamma r_i},$$

$$\kappa_r := (f_{r,\gamma})_{\#} q = \sum_{i=1}^n q_i \delta_{r+\gamma r_i}.$$

It follows from the closed form of MMD that

$$\begin{aligned} \text{MMD}^2(\mathcal{T}^\pi \mu(s_0), \mathcal{T}^\pi \nu(s_0); k) &= \text{MMD}^2(\mathbb{E}_r[\eta_r], \mathbb{E}_r[\kappa_r]; k) \\ &= \sum_{r, r' \in \text{dom}(R)} p_r p_{r'} \text{MMD}^2(\eta_r, \kappa_{r'}; k) \\ &> \sum_{i=1}^n p_{r_i}^2 \text{MMD}^2(\eta_{r_i}, \kappa_{r_i}; k) \\ &\stackrel{(a)}{=} \left(\sum_{i=1}^n p_{r_i}^2 \right) \text{MMD}^2(\eta_0, \kappa_0; k) \\ &\stackrel{(b)}{=} \gamma^{2\alpha} \text{MMD}^2(\eta_0, \kappa_0; k). \end{aligned} \tag{4}$$

Here (a) follows from that Gaussian kernels are shift invariant and (b) follows from the particular choice of ϵ such that $\sum_{i=1}^n p_{r_i}^2 = \gamma^{2\alpha}$. Similarly, we have

$$\text{MMD}^2(\mathcal{T}^\pi \mu(s_1), \mathcal{T}^\pi \nu(s_1); k) > \gamma^{2\alpha} \text{MMD}^2(\eta_0, \kappa_0; k).$$

It remains to choose particular values of $\sigma, n, p, q, \{r_i\}, \gamma$ such that $\gamma^2 \geq \frac{1}{n}$ and $\text{MMD}^2(\eta_0, \kappa_0; k) \geq \text{MMD}^2(p, q; k)$. It is indeed possible by choosing the values as in Table 2.

So far, we have constructed a particular instance such that

$$\begin{aligned} \text{MMD}(\mathcal{T}^\pi \mu(s_0), \mathcal{T}^\pi \nu(s_0); k) &> \gamma^\alpha \text{MMD}(\mu(s_0), \nu(s_0); k) \\ \text{MMD}(\mathcal{T}^\pi \mu(s_1), \mathcal{T}^\pi \nu(s_1); k) &> \gamma^\alpha \text{MMD}(\mu(s_1), \nu(s_1); k). \end{aligned}$$

Thus, we have

$$\text{MMD}_\infty(\mathcal{T}^\pi \mu, \mathcal{T}^\pi \nu; k) > \gamma^\alpha \text{MMD}_\infty(\mu, \nu; k),$$

which contradicts Inequality (3).

Case 2: For exp-prod kernels $k(x, y) = \exp(xy/\sigma^2)$.

We follow the same procedure as in Case 1 but only up to Eq. (4) as the exp-prod kernel is not shift invariant. Instead, define

$$r^* = \arg \min_{r \in \{r_1, \dots, r_n\}} \text{MMD}(\eta_r, \kappa_r; k).$$

Then, we have

$$\begin{aligned} \text{MMD}^2(\mathcal{T}^\pi \mu(s_0), \mathcal{T}^\pi \nu(s_0); k) &> \sum_{i=1}^n p_{r_i}^2 \text{MMD}^2(\eta_{r_i}, \kappa_{r_i}; k) \\ &\geq \left(\sum_{i=1}^n p_{r_i}^2 \right) \text{MMD}^2(\eta_{r^*}, \kappa_{r^*}; k) \\ &= \gamma^{2\alpha} \text{MMD}^2(\eta_{r^*}, \kappa_{r^*}; k). \end{aligned}$$